EARLY-STAGE RESULTS

Emotion-Mediated Personalized Image Aesthetic Assessment

Predicting individual aesthetic preferences through emotional responses

PIAA EMOTION MEDIATION XPASS-VIS

Presenter: Pinwa | Date: June 14, 2026

02 — INTRODUCTION

What is Beauty?

AESTHETIC SUBJECTIVITY

The Subjectivity of Aesthetic Assessment

Beauty is not a single objective score. It is a multi-dimensional construct shaped by personal experiences, cognitive interpretations, and emotional resonance.

Emotional Resonance

Human judgment of beauty is closely bound to emotional reactions. A stunning mountain sunset evokes feelings of awe, calm, or nostalgia.

Cognitive Variations

Individuals bring their unique cognitive backgrounds, personal experiences, and preferences to every viewing, leading to wide variations in ratings.

Beyond Numeric Regressions

Conventional models simplify beauty into a single number. Modern assessment requires capturing the emotional mediators behind that judgment.

The Core Problem.

The exact same image receives completely different beauty scores from different people.

Subjectivity of Beauty

Beauty is in the eye of the beholder. If we show the same picture to different observers, their responses exhibit high variance. Predicting a single average rating fails to capture individual preference.

What is PIAA?

Personalized Image Aesthetic Assessment (PIAA) is a specialized field in AI. Instead of predicting a generic general aesthetic score, the goal is to train an AI model capable of predicting the specific aesthetic score that a **particular individual** would assign to an image.

Our Ultimate Goal:

Predict the score for one specific person

04 — METHODOLOGY

Our Proposed Core Idea.

TRADITIONAL METHODS

Direct Prediction (Black Box)

Image → Deep AI → Beauty Score

Most previous methods map the image directly to a personal aesthetic score. It operates like a black box: we do not know why the model predicted that specific rating, and it lacks transparency.

OUR APPROACH

EXPLAINABLE

Emotion-Mediated Prediction

Image → Emotions → Formula → Personal Score

We introduce an explicit intermediate step: **predicted emotions**. Each user is modeled by a **personal formula** (emotion weights). The aesthetic score is computed as a weighted sum of these emotions, making the prediction fully interpretable.

Grounded in Psychology Research

Our method is based on human psychological pathways: people do not judge beauty directly from raw image pixels. First, the image triggers an emotional response (e.g. nostalgic, amused, or impressed), and that feeling serves as the foundation for the aesthetic judgment.

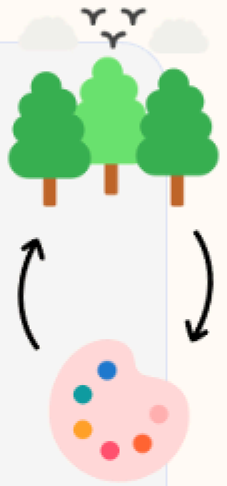
05 — CORE HYPOTHESES

Research Questions.

Q1 Accuracy vs. Direct Prediction

Can emotion-mediated modeling perform as accurately as direct aesthetic prediction?

Comparing our explainable two-stage pipeline against traditional black-box deep models to see if we sacrifice performance for transparency.



Q2 Cross-Category Consistency

Is a person's "personal emotion formula" consistent across different image categories?

If a user appreciates fine art due to nostalgia, do they use the same criteria for fashion or landscape photos, or does their formula shift?

Q3 Psychological Grounding

Does the personal formula correlate with real-world personality traits?

Testing if users high in Openness to Experience show stronger weights for 'Intellectually Challenged' in their formula, confirming psychological grounding rather than random data fitting.

Q4 Mediation Redundancy

Should 'Liked' or 'Beautiful' be excluded as mediating emotions?

Since these dimensions are close to the final aesthetic rating, does including them make prediction trivial? We report on both 9-emotion and 7-emotion models to ensure robustness.

06 — DATA SOURCE

The XPASS-Vis Dataset.

IMAGES
6,526
Art, Fashion, Landscape

PARTICIPANTS
129
Individual evaluators

TOTAL RATINGS
87,836
Fully labeled examples

LABELS PER RATING
1 + 9
1 Beauty (1-7) + 9 Emotions (1-5)

Note: Created by Hayashi-san's lab. The inclusion of granular, multi-dimensional emotional ratings per person is what makes our personalized approach uniquely possible. Most traditional datasets only contain general aesthetic averages.

List of 9 Rated Emotions

POSITIVE

- Liked it
- Found it beautiful
- Impressed
- Motivated
- Amused

COGNITIVE

- Nostalgic
- Intellectually challenged

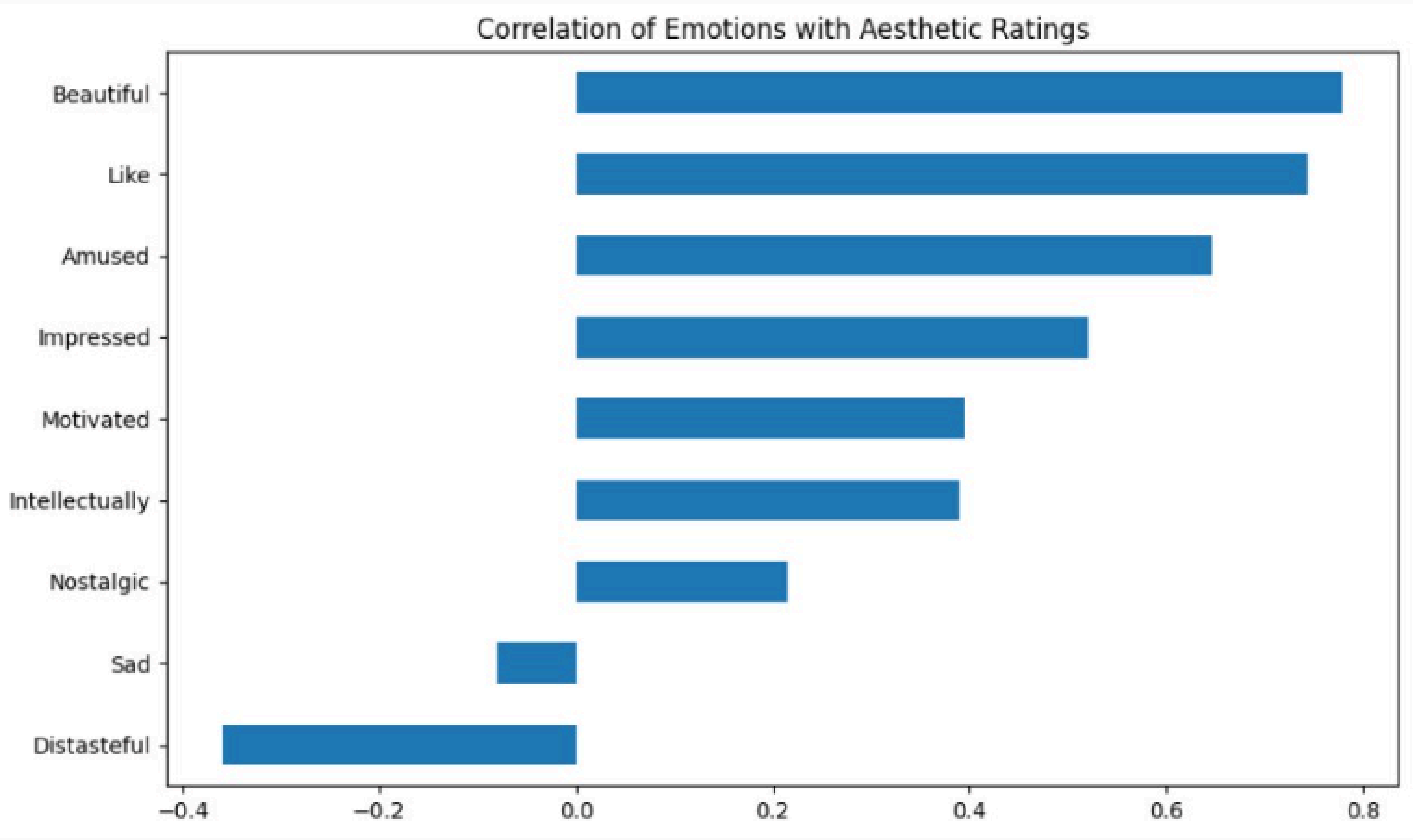
NEGATIVE

- Sad
- Distasteful

07 — DATA EXPLORATION

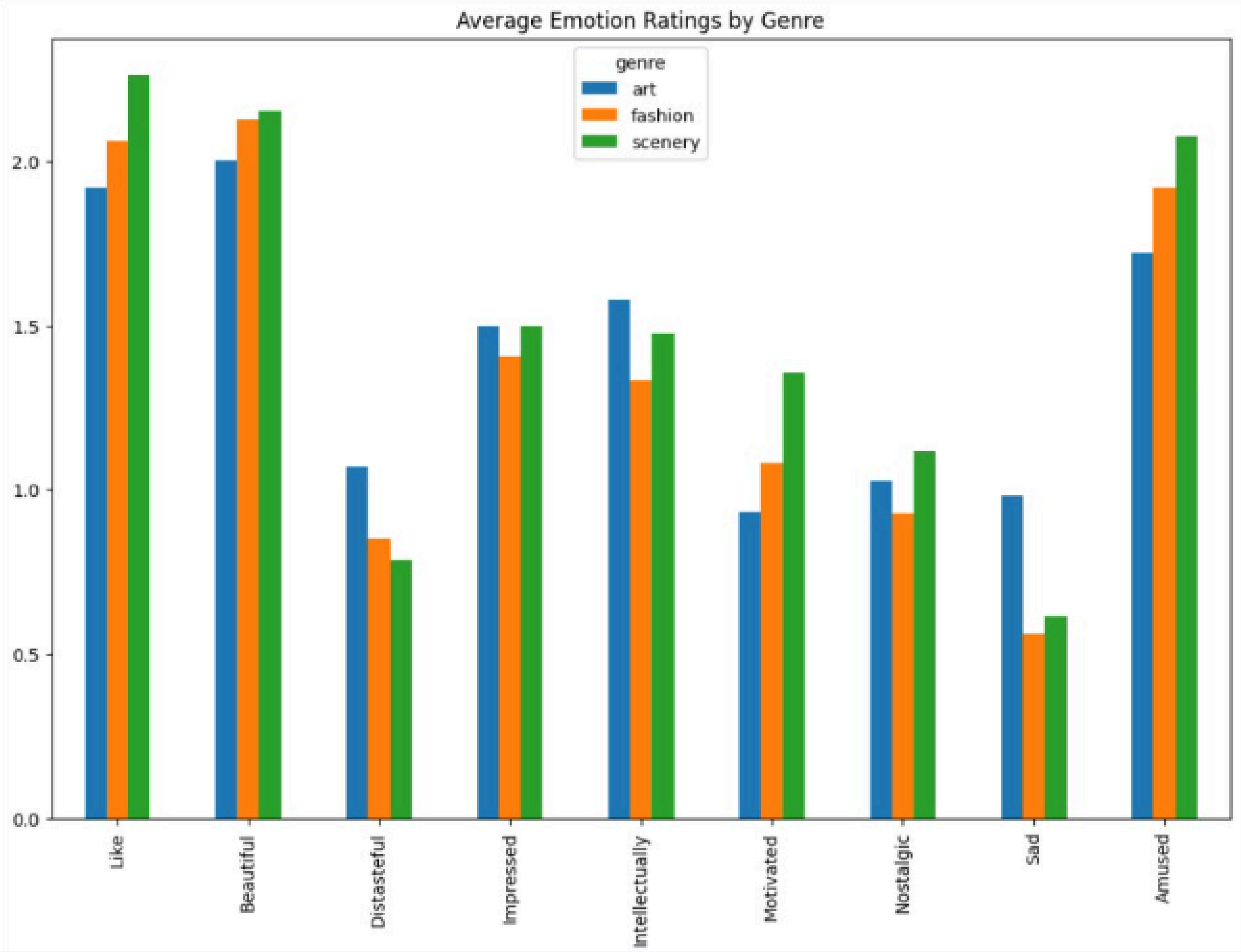
Dataset Visualizations.

Emotion Correlations with Aesthetics



Positive emotions like 'Beautiful' and 'Like' show high correlation with overall aesthetics, while negative emotions like 'Distasteful' are highly negatively correlated.

Average Emotion Ratings by Genre



Aesthetic ratings vary significantly across different genres (Art, Fashion, Scenery), with positive emotions scoring higher on average for Scenery images.

08 — ARCHITECTURE

The Prediction Pipeline.

Input Image

Any artwork, fashion photograph, or landscape scene.

TRAINED ON ALL USERS

Step 1: Shared AI

A generic model that predicts the 9 emotional response levels for the image (e.g. nostalgia = 4/5).

9 WEIGHTS PER USER

Step 2: Personal Formula

Each person has a unique set of weights representing how much they care about each emotion when judging beauty.

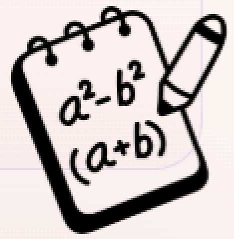
Personal Score

A custom aesthetic prediction tailored to that specific user.

THE PERSONAL FORMULA

The formula simply evaluates: **How much does this specific person care about each of the 9 emotions when deciding if an image is beautiful?**

Because these weights are explicit, we avoid the black-box issue. We can look at the 9 numbers and read exactly what aesthetic priorities characterize each individual user.



09 — FEASIBILITY ANALYSIS

Experiment 1: Is the Idea Even Possible?

METHOD / DATASET INPUT	ART	FASHION	LANDSCAPE	AVERAGE
All 9 emotions Using the full set of emotional responses	0.819	0.838	0.843	**0.833**
7 emotions (Real emotions) Removed circular proxies: 'liked' & 'beautiful'	0.716	0.705	0.740	**0.720**
Best previous method (ICI) Shi et al., 2024 (without emotions)	0.493	0.319	0.458	**0.423**

** Values represent Concordance Correlation Coefficient (CCC) score, max = 1.0 (higher is better).*

Theoretical Ceiling: 0.720

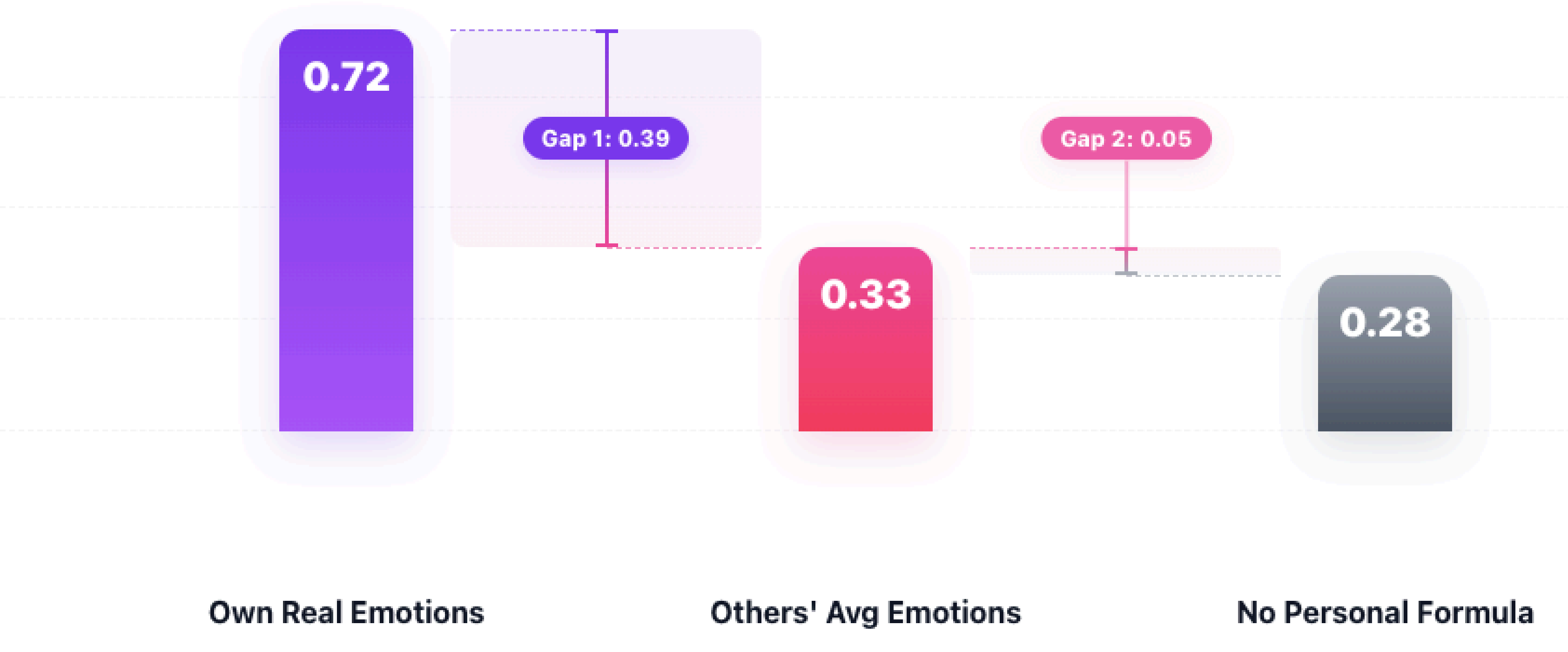
Evaluating with Circular Proxies

Using all 9 emotions yields a prediction score of **0.833**. However, emotions like "liked it" and "found it beautiful" are circular references for aesthetic preference.

The Real Emotional Mediation

Removing circular proxies leaves 7 "pure" emotions. This still achieves a high score of **0.720**, nearly doubling the previous best method (**0.423**). This proves the feasibility of the concept.

Experiment 2: Where does Personalization Come From?



GAP 1 — FEELING DIFFERENTLY (0.39)

This represents observer subjectivity in emotional response. Different people look at the exact same image and feel completely different emotions (e.g. nostalgic vs neutral). This accounts for **88%** of the personalization drop.

GAP 2 — CARING DIFFERENTLY (0.05)

This represents observer difference in weightings. Even if two people feel the exact same emotions, they value them differently when assessing overall beauty. This accounts for only **12%** of the difference.

Crucial Insight: Personalization comes mainly from **how people feel about an image**, not how they weigh those feelings. This finding is completely unique to the XPASS-Vis dataset.

Experiment 3: Can We Predict Individual Emotions?

MODEL METHOD	SUPPORT = 10	SUPPORT = 25	SUPPORT = 50	SUPPORT = 100
Direct (No Emotion) Predicting aesthetic score directly from pixels	0.109	0.187	0.245	0.294
Through Emotion + Correction (Hybrid) Proposed emotion-mediated route	0.153	0.238	0.299	0.334
Improvement Relative performance gain	+40%	+27%	+22%	+14%

* Support represents the number of images rated by the person used to train their personalized formula.

ICI State-of-the-Art: 0.423 **Theoretical Ceiling:** 0.720

🚀 Highest Gain in Scarce Data

The improvement is largest (**+40%**) when we have the least data (only 10 images per user). This highlights how using emotional mediation acts as a strong prior, preventing overfitting.

📈 Consistent Outperformance

Going through predicted emotions and applying correction consistently beats direct score prediction at all support levels (10 to 100 images).

🔍 Room for Improvement

We are currently at **0.334**, which is below ICI (**0.423**). However, the theoretical ceiling is **0.720**, indicating that refining the emotion predictor offers a huge path forward.

12 — CONTEXT

Literature & Related Work.

PAPER	CONTRIBUTION	RELATION TO OUR WORK
ICI (Shi et al., 2024)	Predicts personal aesthetic score directly using image features and user metadata.	Ours uses emotions as an explainable intermediate step rather than a black-box end-to-end mapping.
HNEF (Lan et al., 2024)	Leverages emotional attributes as supplementary features to improve general (non-personal) aesthetic predictions.	Ours focuses on personalization, using emotional reactions as the primary signal to capture individual preference.
Iigaya et al. (2021, Nature Human Behaviour)	Demonstrated that human aesthetic preferences can be modeled as a weighted linear combination of visual features.	Serves as the theoretical backing for our 'personal formula' concept (modeling beauty as a weighted sum of emotions).

Key Takeaway: Our approach is not just an arbitrary engineering heuristic. By grounding the "personal formula" in the weighted-feature preference model established by **Iigaya et al. (2021)**, we align our AI architecture directly with how psychologists believe human aesthetic evaluation operates.

Research Challenges.

Predicting Personal Emotions

An image can make different people feel very different things. A general AI model typically only learns the average human reaction, not individual emotional subjectivity.

Rare Emotion Sparsity

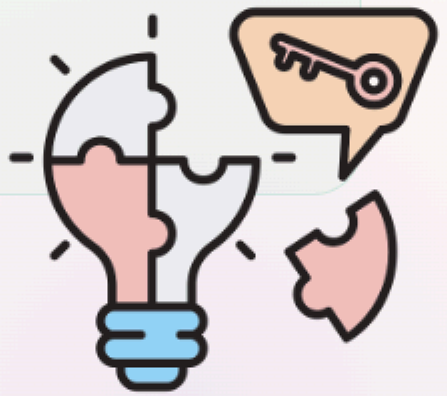
Certain emotions occur very rarely in particular domains, causing data imbalance that is hard to train.

Limited Personalized Data

Each participant rated ~100 images. Fitting the 9-dimensional personal formula on such limited samples poses a significant risk of overfitting.

Evaluation Alignment

To compare fairly against existing baselines, we need identical cross-validation data splits across all models.



14 — FUTURE DIRECTIONS

Next Research Steps.

01

Complete 5-Fold Cross-Validation

Run comprehensive experiments across all remaining folds to secure robust, stable benchmark metrics against ICI.

02

Correlate with Personality (Big Five)

Cross-reference each person's emotion weightings (personal formula) with their Big Five personality profiles already present in XPASS-Vis.

03

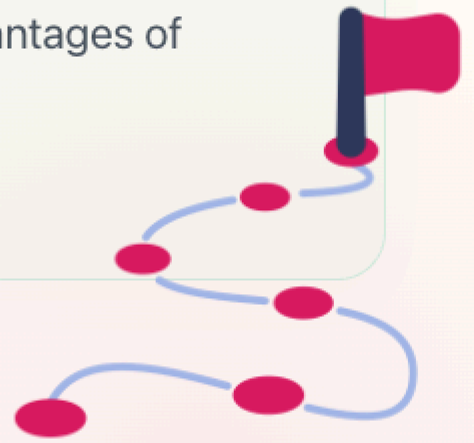
Test Domain Consistency

Analyze whether an individual's personal formula stays consistent when evaluating different domains (e.g. switching from landscape to fashion).

04

Paper Write-up & Submission

Synthesize findings, document the explainability advantages of emotion mediation, and start drafting the paper.



Thank You!

Happy to take any questions or suggestions.